Week 07: ER modelling and relational mapping

Learning outcomes: CLO1, CLO2

Lesson

- Design steps: **conceptual** (things and links) → **logical** (tables, keys) → **physical** (types, indexes, one DBMS).
- **Entity** = thing (province). **Attribute** = property (name). **Relationship** = link (province has districts).
- **Cardinality:** 1:1, 1:N, M:N. **Participation:** is the link required?
- 1:N → FK on the "many" side. M:N → junction table.
- **Weak entity:** identified through its owner. **Specialization:** subtypes of an entity. Our data has only 1:N, so show the others on paper.
- Draw the ERD in DBML and view it on dbdiagram.io.

Worked example

Run `example.sql` (how (`setup/README.md`)), then answer in `schema.dbml` and `work.sql`.

Tables for this week

Read `cambodia`, write `student_work`. All columns are `NOT NULL`. Samples are the first two rows by code.

`cambodia.province`

Column	Type	Key	Meaning
`province_pcode`	`CHAR(4)`	PK	Province code
`province_name`	`VARCHAR(100)`		Province name (English)
`area_km2`	`DECIMAL(12,1)`		Area in km², one decimal place

Sample rows:

province_pcode	province_name	area_km2
KH01	Banteay Meanchey	6145.9
KH02	Battambang	11869.0

`cambodia.district`

Column	Type	Key	Meaning
`district_pcode`	`CHAR(6)`	PK	District code
`district_name`	`VARCHAR(100)`		District name (English)
`area_km2`	`DECIMAL(12,1)`		Area in km², one decimal place
`province_pcode`	`CHAR(4)`	FK	Province code

Sample rows:

district_pcode	district_name	area_km2	province_pcode
KH0102	Mongkol Borei	807.2	KH01
KH0103	Phnum Srok	761.7	KH01

cambodia.commune

Column	Type	Key	Meaning
commune_pcode	CHAR(8)	PK	Commune code
commune_name	VARCHAR(100)		Commune name (English)
area_km2	DECIMAL(12,1)		Area in km², one decimal place
district_pcode	CHAR(6)	FK	District code

Sample rows:

commune_pcode	commune_name	area_km2	district_pcode
KH010201	Banteay Neang	52.0	KH0102
KH010202	Bat Trang	40.3	KH0102

Join keys: `district.province_pcode = province.province_pcode`; `commune.district_pcode = district.district_pcode`.

More: data guide (`data/README.md`).

Lab tasks

1. Complete `schema.dbml`: three tables, columns, PKs and FKs.
2. Mark each relationship 1:N and show that every child needs a parent.
3. Check with SQL that every district and commune has a parent.
4. Check whether every province and district has a child. Data today vs what the schema allows?
5. Write CREATE TABLE statements in `student_work` from your ERD, with the source data types.

Homework

1. Explain natural vs surrogate keys using P-codes and MongoDB `_id`.
2. Describe conceptual, logical and physical design for this data. Sketch an M:N junction, a weak entity and a specialization on paper.

Capstone sprint

C7: Submit ERD v1 and a data dictionary; justify keys and relationships.

Submit

Answers in `work.sql`, reflection in `reflection.md`, output in `results/`. Commit and push. See SUBMISSIONS.md.